

NovaArcade

Product Requirements Document | NovaArcade Web Platform

Stack	HTML5, CSS3, Vanilla JavaScript (ES6+)
Target	Responsive web browser (desktop, tablet, mobile)
Hosting	Static files / local folder, no backend server
Primary Data File	data/games.json

1. Project Overview

NovaArcade is a multi page, browser based gaming portal that lets visitors browse a library of casual games, view details for a single game, play it directly inside the page through an embedded player, and track their personal best scores on a local leaderboard. The site requires no backend server and no build tooling: it runs by opening index.html in a browser or by serving the folder as static files. All game metadata is sourced from a single games.json file, and all user specific data (scores, favourites, display name) is persisted in the browser's localStorage under the novaarcade_ key prefix.

2. Technology Requirements

Core Stack

The site is built using plain HTML5, CSS3, and vanilla JavaScript (ES6 or later). Frontend frameworks such as React, Vue, or Angular are not used, and no build tools, bundlers, or transpilers are required.

Allowed External Libraries

Library	Purpose
Google Fonts	Loads the Orbitron and Inter font families
Font Awesome	Icon set used for navigation and buttons

Data Storage

Store	Location / Key	Contents
Game catalogue	data/games.json	id, title, category, thumbnail, embedUrl, description, instructions, dateAdded, playCount
User scores	localStorage: novaarcade_score_	Highest recorded score per game
Favourites	localStorage: novaarcade_favourites	JSON array of favourited game ids
Display name	localStorage: novaarcade_username	Locally saved profile name

3. Site Structure

The site is composed of four HTML pages, each with a single responsibility, linked together through a shared navigation bar.

Page	Role
index.html	Home page. Displays the hero banner, the Featured Games row, the New Releases row, and category quick links.

Page	Role
library.html	Full game library as a grid of cards, with the search bar and category filter sidebar.
game.html	Detail view for a single game, selected via an id query parameter (e.g. game.html?id=neon-runner). Shows description, instructions, embedded player, and related games.
leaderboard.html	Local leaderboard table showing the player's best score per game across all scorable games.

Shared Components

Component	Location	Contents
Navigation Bar	Top of every page	Site logo, links to Home / Library / Leaderboard, and a search icon
Footer	Bottom of every page	Site name, links labelled About / Contact / Terms, and a copyright line

4. Home Page

The home page is composed of four sections, displayed in a fixed top to bottom order: the hero banner, the Featured Games row, the New Releases row, and the category quick links.

Section	Contents	Behaviour
Hero Banner	Full width background image, heading "Play Free Games Instantly", one line subheading, "Browse Games" button	Button links to library.html
Featured Games	Exactly 4 game cards (thumbnail, title, category tag, "Play Now" button)	"Play Now" links to game.html?id=
New Releases	Exactly 4 game cards, same layout as Featured Games	Cards correspond to the 4 games.json entries with the most recent dateAdded values
Category Quick Links	One badge per category present in games.json	Clicking a badge opens library.html?category= with that category pre selected

5. Game Library Page

Layout

The page uses a two column layout: a filter sidebar on the left and a game grid on the right. On screens narrower than 768px the sidebar collapses into a toggleable panel (see Section 12).

Filter Sidebar Controls

Control	Type	Behaviour
Category Filter	Checkbox list	One checkbox per category found in games.json. Checking a box adds that category to the active filter set.
Sort Order	Dropdown select	Options: "Newest First", "Most Played", "Alphabetical"
Clear Filters	Button	Unchecks all category checkboxes and resets the sort dropdown to "Newest First"

Game Grid

Each card displays the thumbnail image, title, category tag, and a play count formatted as "X plays" (X = playCount from games.json). Clicking a card navigates to `game.html?id=`. The grid shows only games that match the active category filters and search term. If no games match, the grid area displays the text "No games match your filters", centred.

6. Game Detail Page

The page reads the id query parameter from the URL and locates the matching entry in games.json. If no entry matches, the page displays the text "Game not found" and a "Back to Library" button linking to library.html. Otherwise the page is populated with the matched entry's data, in the following section order.

Section	Contents
Title Bar	Game title as a heading, with the category tag beside it
Embedded Game Player	As specified in Section 7
Description Panel	Full text of the description field
How to Play Panel	Instructions field rendered as a numbered list, one list item per line
Related Games Row	Up to 4 other games.json entries sharing the same category, using the home page card layout. Fewer than 4 shown if fewer exist.

7. Embedded Game Player

Player Container

Each games.json entry includes an embedUrl pointing to a publicly playable HTML5 game. The detail page embeds this URL inside an iframe held in a fixed 16:9 aspect ratio container. A loading placeholder showing the text "Loading game..." centred over a solid background colour is displayed until the iframe's load event fires, then hidden.

Score Capture

Trigger	Behaviour
Embedded game sends postMessage with a score field	Value is compared against the existing localStorage entry novaarcade_score_
New score is higher than the stored score	Stored value is updated and the leaderboard data (Section 11) is updated
New score is not higher	No update occurs

Fullscreen Control

A button labelled "Fullscreen" appears below the iframe. Clicking it calls requestFullscreen on the iframe element.

8. Search and Filter System

Behaviour	Detail
Search expand	Clicking the navigation bar search icon expands a text input field
Search submit	Pressing Enter navigates to library.html with the entered text in a query parameter named q
Title match	On library.html, if q is present the grid shows only games whose title contains q, case insensitive
Combined filtering	When both q and one or more category checkboxes are active, a game must satisfy both conditions to appear in the grid

9. User Account Panel

Local Profile

A "Sign In" button in the navigation bar opens a modal dialog containing a text input labelled "Display Name" and a "Save" button. Clicking "Save" stores the entered name in localStorage under novaarcade_username and closes the modal. Once a name is saved, the "Sign In" button is replaced with the saved name; clicking it reopens the same modal pre filled with the current name for editing.

Favourites

On the library page and the game detail page, every game card and the detail page title bar include a heart shaped icon button. Clicking it toggles the game's id in the JSON array stored under novaarcade_favourites. The icon renders filled when the id is present in the array and outline when absent.

10. Leaderboard System

Aspect	Specification
Data source	All localStorage keys beginning with novaarcade_score_; game titles resolved from games.json via the id portion of each key
Table columns	"Game", "Your Best Score", "Category", in that order
Row order	Descending by score value
Empty state	If no novaarcade_score_ keys exist, display "No scores yet. Go play some games!" with a "Browse Games" button linking to library.html, in place of the table

11. Visual Style

Colour Palette

Name	Hex	Usage
Background Dark	#12121A	Page background
Surface	#1E1E2E	Card and panel backgrounds
Primary Accent	#7B5CFF	Buttons, active links, highlights
Secondary Accent	#00E0C7	Hover states, badges
Text Primary	#F5F5FA	Main text colour
Text Muted	#9A9AB0	Secondary text, captions

Typography and Buttons

Headings use the Orbitron font family. Body text, list items, and form labels use the Inter font family. Both are loaded from Google Fonts. Primary action buttons ("Play Now", "Browse Games", "Save") use the Primary Accent colour as background with Text Primary as the font colour and 8px rounded corners. On hover, the background transitions to the Secondary Accent colour over 0.2 seconds.

12. Responsive Behaviour

Breakpoint	Width Range	Behaviour
Desktop	1024px and above	Full multi column layouts as specified per section
Tablet	768px to 1023px	Game grids reduce to 2 columns; Featured and New Releases rows scroll horizontally
Mobile	Below 768px	Game grids reduce to 1 column; the library filter sidebar collapses behind a "Filters" toggle; navigation links collapse into a hamburger menu

On mobile widths, the navigation bar shows a hamburger icon on the right. Clicking it slides a vertical menu panel in from the right edge containing the desktop links stacked vertically. Clicking the icon again, or clicking outside the panel, closes it.

13. Code Quality and File Structure

Minimum file structure:

```
index.html
library.html
game.html
leaderboard.html
css/
  styles.css
js/
  main.js
  library.js
  game.js
  leaderboard.js
data/
  games.json
assets/
  images/
```

- Shared logic (card rendering, localStorage access, navigation bar rendering) lives in main.js and is reused by the page specific scripts; logic is not duplicated across files.
- All JavaScript avoids global variable pollution by wrapping page logic in functions or modules.
- All CSS uses custom properties (CSS variables) for the Section 11 colour palette, declared once in the :root selector of styles.css.
- Images in assets/images are jpg, png, or webp and do not individually exceed 500KB.
- The lazy loading attribute is applied to all thumbnail images that appear below the first visible section of any page.

14. Scope Boundaries

The following are explicitly out of scope:

- User authentication beyond the local display name in Section 9. No passwords, accounts, or server side sessions.
- Payment, purchase, or in app currency systems of any kind.
- Any backend server, database, or API beyond reading the static games.json file.
- Multiplayer or real time features.

- Building games from scratch; all games are embedded via the Section 7 iframe mechanism using existing publicly playable HTML5 games.
- Analytics, tracking scripts, or third party advertising integrations.

15. Deliverables

All 12 items are required. No deliverable may be partial or omitted.

ID	Deliverable	Type	Notes
D-01	index.html with hero banner, Featured row, New Releases row, category quick links	Page	Must match Section 4
D-02	library.html with filter sidebar, sort dropdown, and game grid	Page	Must match Section 5
D-03	game.html with title bar, embedded player, description, instructions, related games	Page	Must match Section 6
D-04	leaderboard.html with sorted score table and empty state	Page	Must match Section 10
D-05	Shared navigation bar and footer on all four pages	Component	Identical markup and styling across pages
D-06	games.json with at least 12 entries across at least 3 categories	Data	Each entry includes id, title, category, thumbnail, embedUrl, description, instructions, dateAdded, playCount
D-07	Working search bar with query parameter based filtering	Feature	Section 8
D-08	Working category and sort filters on the library page	Feature	Section 5
D-09	Local sign in modal with persistent display name	Feature	Section 9
D-10	Favourites toggle with persistent heart icon state	Feature	Section 9
D-11	Score capture from embedded games via postMessage and leaderboard update	Feature	Sections 7 and 10
D-12	Responsive layout across desktop, tablet, and mobile breakpoints, including hamburger menu	Styling	Section 12

16. Technical Notes for the Development Team

- All colour values, breakpoint widths, and category lists are driven by the values in this document and in games.json.
- No placeholder sample text (e.g. "Lorem ipsum") may remain in the final delivery.
- Every interactive element described in this document, including buttons, toggles, and modals, must be functional, not only styled.